

AgriHGT: Heterogeneous Graph Transformer for Multi-Crop Leaf Disease Classification

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Abstract—Leaf disease classification in multi-crop field images is usually treated as independent image recognition, although crops, diseases, and disease categories are related and visually similar leaves carry complementary evidence. We propose AgriHGT, a two-stage visual-relational framework that places partially fine-tuned backbone features of each leaf image in a heterogeneous graph with crop, disease, and disease-category nodes, propagates information with a Heterogeneous Graph Transformer, fuses the result with a gated residual visual path, and predicts the disease with a leaf-disease compatibility decoder. On the 28-class Agri-Vision4 dataset (Tomato, Papaya, Zucchini, Bottle Gourd), we build a duplicate-aware, leakage-controlled split through MD5 deduplication, perceptual-hash linking, and an iterative near-duplicate audit, and prevent label shortcuts by separating message-passing and supervised training nodes. Seven convolutional, transformer, and self-supervised backbones, three message-passing depths, a feature-only logistic-regression reference, and a nine-configuration component ablation are evaluated. The best configuration, [backbone, depth], reaches [xx.xx]% test accuracy and [0.xxxx] macro-F1, compared with [xx.xx]% for the corresponding feature-only reference, and the ablation shows that [component] contributes most.

Index Terms—Plant disease classification, heterogeneous graph transformer, graph neural network, self-supervised learning, DINOv3, Agri-Vision4.

I. INTRODUCTION

Leaf diseases reduce vegetable yield, and in smallholder settings such as Bangladesh expert diagnosis is often unavailable. Deep learning on leaf images is the standard route to automated diagnosis [1], [2], but it classifies each image independently. In a multi-crop setting, a leaf belongs to a crop, its disease to a broader category spanning crops, and visually similar leaves tend to share a diagnosis, structure that image-only classifiers ignore. Graph models relate regions or leaves [3], [4] and knowledge graphs encode crops and diseases [5]–[7], yet they are rarely combined, and combining them risks augmented copies leaking across partitions, labels returning to supervised nodes through the graph, and conclusions valid for one backbone only.

We propose AgriHGT, which joins image features and agricultural knowledge in one heterogeneous graph processed by the Heterogeneous Graph Transformer (HGT) [8]. Contributions: (1) a 26,905-node, 13-relation graph linking leaf images to crops, diseases, and categories, with gated residual fusion and a compatibility decoder; (2) a leakage-controlled protocol for Agri-Vision4 [9] (exact deduplication, perceptual-hash linking, iterative near-duplicate audit, per-class support

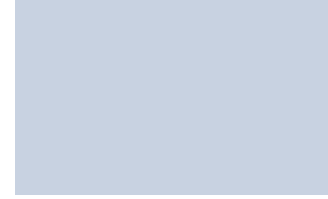


Fig. 1. Overview of the AgriHGT methodology.

control) plus a label-shortcut-free message/supervision partition; (3) a comparison of seven backbones, three HGT depths, and a feature-only reference; and (4) a nine-configuration ablation with McNemar tests and class-balanced, chance-corrected, and calibration metrics.

II. RELATED WORK

Image-based recognition. CNN transfer learning [1], [2], [10]–[12], vision and Swin transformers [13]–[15], CNN-ViT and attention-based multi-crop models [16]–[18], and self-supervised learning [19], [20] all treat images independently.

Graphs and knowledge. Graph models relate visual regions or leaf images [3], [4], and agricultural knowledge graphs link crops, diseases, pests, and symptoms [5]–[7]. HGT’s type-specific parameters and relation-aware attention [8] let AgriHGT place images and knowledge entities in one graph.

III. MATERIALS AND METHODS

A. Overview

AgriHGT is a two-stage visual-relational classifier (Fig. 1). First, a pretrained visual backbone is partially fine-tuned on the training partition and used to extract one fixed feature vector per retained leaf image. Second, these features become `leaf_image` nodes in a heterogeneous graph with crop, disease, and disease-category nodes, on which relation-aware message passing is performed with the Heterogeneous Graph Transformer (HGT) [8]. The graph representation is fused with a residual visual representation before disease prediction.

B. Dataset

Agri-Vision4 [9] contains leaf images of Tomato, Papaya, Zucchini, and Bottle Gourd in 28 crop-specific healthy and disease classes, collected in Bangladeshi fields under natural light and validated by agricultural experts. It provides original images and augmented copies (rotation, flipping, brightness and contrast adjustment, noise); augmented images form a large share of every crop (Fig. 2). Before duplicate removal

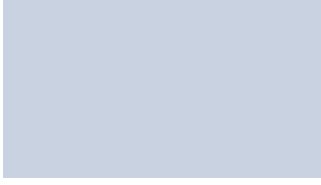


Fig. 2. Original and augmented images across the four crops of Agri-Vision4.

there were 28,121 files, of which 5,246 were originals (Table I). Because class frequencies are unequal, accuracy is reported with macro-F1 and other class-balanced metrics.

C. Leakage-Controlled Data Partitioning

Exact duplicates. MD5 hashing of all 28,121 files removed 1,255 byte-identical duplicates, leaving 26,866 images (4,304 original, 22,562 augmented).

Duplicate-aware initial split. To keep an image and its transformed copies together, originals were split first (class-stratified 70/15/15: 3,004/650/650). Perceptual hashes with `hash_size=16` (256 bits) were then computed, and each augmented image was compared with originals of the same crop and class; if the minimum Hamming distance was at most six, it was linked to the closest original and assigned to its partition. This linked 1,552 augmented images; the other 21,010 could not be confidently associated and were assigned to training.

Near-duplicate audit. Same-class perceptual hashes were then compared across partitions for original–original, original–augmented, and augmented–augmented pairs. Whenever a cross-partition pair had Hamming distance ≤ 6 , the validation or test image was moved to training, repeating until no violation remained; successive rounds found 41, 11, 3, and 0 violating pairs.

Per-class support control. A class left without a validation or test image received an eligible training image, re-audited and swapped if it created a new near-duplicate relation. The final split has 25,122 training, 896 validation, and 848 test images, with at least one validation and one test image per class, and is shared by all backbone experiments. It is *leakage-controlled* rather than *leakage-free*: the audit covers same-class pairs and the chosen hash and threshold, so the supported claim is that no same-class cross-partition pair exists at Hamming distance ≤ 6 .

D. Visual Backbones

Backbone features initialize `leaf_image` nodes and define `similar_to` edges, so the graph changes with the backbone. Seven backbones spanning convolutional, transformer, supervised, and self-supervised families used in recent plant-disease studies [16]–[18] were compared (Table II), both to find the strongest one and to test whether heterogeneous graph learning behaves consistently across representations. ResNet50 is the standard residual CNN reference used in plant-disease transfer learning [10], [11]; DenseNet121’s dense connections reuse features so small lesions and larger textures both stay available; EfficientNet-B3’s compound scaling suits

multi-scale symptoms [12]; ConvNeXt-Tiny, a transformer-inspired CNN, tests whether gains need attention; and Swin-Tiny adds shifted-window hierarchical attention [13]–[15]. The self-supervised DINOv2 ViT-B/14 and DINOv3 ViT-B/16 learn from unlabeled images an embedding that may preserve structure beyond supervised classes, which matters because the graph needs meaningful neighborhoods, not only a strong classifier [19], [20]; comparing them isolates the effect of representation generation within one family.

E. Backbone Adaptation and Feature Extraction

Because generic features may emphasize shape and background over subtle color and texture cues, each backbone was first adapted on the training partition with a temporary 28-class head, fine-tuning only the last stage: `layer4` for ResNet50; `denseblock4`, `norm5`, and `classifier` for DenseNet121; `features[7:]` for EfficientNet-B3; `features[7]` for ConvNeXt-Tiny; the final stage and terminal norm for Swin-Tiny; and the final two blocks and terminal norms for DINOv2 and DINOv3, each with its head. Partial fine-tuning limits cost and preserves low- and mid-level features. Validation accuracy was monitored every epoch (with patience-based early stopping where used), the best checkpoint was restored, the head was replaced by an identity, and the frozen backbone ran one deterministic pass without augmentation over all 26,866 images at its native resolution and normalization. This gives feature matrices of $26,866 \times d$ with $d = 2,048$ (ResNet50), 1,024 (DenseNet121), 1,536 (EfficientNet-B3), and 768 (ConvNeXt-Tiny, Swin-Tiny, DINOv2, DINOv3). Each dimension was standardized as $x' = (x - \mu_{\text{train}}) / \sigma_{\text{train}}$ with statistics from the training partition only, applied unchanged to validation and test, separately per backbone.

F. Heterogeneous Graph Construction

One graph (Fig. 3) per backbone has 26,905 nodes: 26,866 `leaf_image`, 4 `crop`, 28 `disease`, and 7 `disease_category` nodes (Healthy, Viral, Bacterial, Fungal, Pest, Nutrient Deficiency, Other), with categories fixed from class names before training [6], [7]. Leaf nodes take the standardized backbone vector; non-image nodes have no pixels and receive learnable input embeddings.

Six forward relations are used: `grown_on` (`leaf`→`crop`), `affects` (`crop`→`disease`), `is_type_of` (`disease`→`category`), `similar_to` (`leaf`→`similar leaf`), `has_disease` (selected training `leaf`→`disease`), and `co_category` (`disease`→`disease of the same category`). Adding a reverse for each and a self-loop per node gives 13 relation types. For `similar_to`, following feature-space similarity graphs [4], each image links to its five nearest neighbors of the same crop under

$$d_{\cos}(\mathbf{x}_i, \mathbf{x}_j) = 1 - \frac{\mathbf{x}_i^T \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2}, \quad (1)$$

with same-crop search limiting crop-level shape effects, giving 134,330 forward edges per backbone.

Message and supervision partition. Linking every supervised training image to its true disease would leak its

TABLE I
ORIGINAL IMAGES PER CROP-SPECIFIC CLASS BEFORE DUPLICATE REMOVAL (TOTAL 5,246)

Bottle Gourd (1,658)		Papaya (1,684)		Tomato (916)		Zucchini (988)	
Alternaria Leaf Blight	303	Bacterial Blight	183	Downy	57	Angular Leaf Spot	120
Anthrachnose	276	Carica Insect Hole	318	Healthy	288	Anthrachnose	129
Downy Mildew	286	Curled Yellow Spot	538	Mosaic	195	Downy Mildew	153
Early Alternaria L. Blight	179	Healthy Leaf	189	Spot	311	Dry Leaf	67
Fungal Damage Leaf	39	Mosaic Virus	119	White Spot	65	Healthy	88
Healthy	260	Pathogen Symptoms	286			Insect Damage	78
Mosaic Virus	315	Yellow Necrotic Spot Holes	51			Iron Chlorosis Damage	65
						Xanthomonas Leaf Spot	86
						Yellow Mosaic Virus	202

TABLE II
VISUAL BACKBONES AND THEIR ROLE IN AGRIHGT

Backbone	Architecture	Pretrain.	Reason for inclusion
ResNet50	Residual CNN	Sup.	Standard CNN reference
DenseNet121	Dense CNN	Sup.	Feature reuse
EfficientNet-B3	Scaled CNN	Sup.	Compound scaling
ConvNeXt-Tiny	Modern CNN	Sup.	CNN-transformer bridge
Swin-Tiny	Hierarchical ViT	Sup.	Local-window attention
DINOv2 ViT-B/14	ViT	Self-sup.	Strong SSL features
DINOv3 ViT-B/16	ViT	Self-sup.	Recent DINO generation

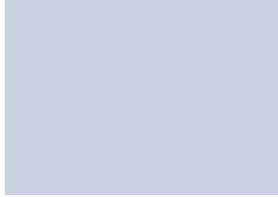


Fig. 3. AgriHGT heterogeneous graph: leaf_image, crop, disease, and disease_category nodes and their typed relations.

label along leaf_image→disease→leaf_image. The training partition was therefore split, class-stratified 75/25 with originals and linked derivatives kept together, into 18,881 train_msg images, which keep has_disease edges but are excluded from the loss, and 6,241 train_sup images, which have no has_disease edge and alone form the loss. Validation and test nodes also lack has_disease edges, so train_sup, validation, and test nodes follow the same rule. With reverse relations and self-loops the graph has 387,301 edges.

G. AgriHGT Architecture

AgriHGT has three components: heterogeneous message passing, gated fusion of graph and visual representations, and a leaf-disease compatibility decoder (Fig. 4).

Projection and HGT. Node features of type $\tau(v)$ are mapped to 256 dimensions, $\mathbf{h}_v^{(0)} = \tanh(W_{\tau(v)}\mathbf{x}_v + \mathbf{b}_{\tau(v)})$. HGT’s type-dependent transformations and relation-aware attention [8] treat similarity and knowledge messages differently. We use 4 heads, hidden size 256, dropout 0.2, and depths $n_{\text{layers}} \in \{0, 1, 2\}$. The zero-layer model keeps the projections, learnable non-image embeddings, residual branch, fusion, and decoder but no message passing; since the original pyHGT code still built one convolution at depth 0, it was modified to perform none.

Gated residual fusion. The standardized image feature is projected as $\mathbf{r}_i = \text{LayerNorm}(W_r\mathbf{x}_i + \mathbf{b}_r)$. With graph-updated leaf representation $\mathbf{h}_i$,

$$\mathbf{g}_i = \sigma(W_g[\mathbf{r}_i \parallel \mathbf{h}_i] + \mathbf{b}_g), \quad \mathbf{z}_i = \mathbf{g}_i \odot \mathbf{r}_i + (1 - \mathbf{g}_i) \odot \mathbf{h}_i, \quad (2)$$

where $\parallel$ is concatenation. The gate bias starts at 2.0 ($\sigma(2) \approx 0.88$), so the model first favors the visual path and learns how much graph information to admit.

Compatibility decoder. Leaf and disease features are projected to 128 dimensions, $\mathbf{u}_i = W_l\mathbf{z}_i + \mathbf{b}_l$ and $\mathbf{v}_j = W_d\mathbf{d}_j$, and scored against all 28 disease nodes:

$$s_{ij} = W_o \text{Dropout}_{0.1}[\text{ReLU}(\mathbf{u}_i + \mathbf{v}_j)] + b_o, \quad (3)$$

and the prediction is $\hat{y}_i = \arg \max_j s_{ij}$, the disease whose representation best matches the leaf.

H. Training and Transductive Setting

Training is full-batch over the whole graph, with loss only on the 6,241 train_sup nodes. Class weights $\tilde{w}_c = 1/n_c$ over train_sup counts are normalized to mean one, and

$$\mathcal{L} = -\frac{1}{|S|} \sum_{i \in S} w_{y_i} \log \frac{\exp(s_{i, y_i})}{\sum_{j=1}^{28} \exp(s_{ij})}. \quad (4)$$

AdamW (learning rate 10^{-3} , weight decay 10^{-4}), gradient clipping at 1.0, and cosine decay toward 10^{-6} are used. ResNet50, Swin-Tiny, and DenseNet121 runs use up to 300 epochs with patience 300 (the full budget, keeping the best validation checkpoint); ConvNeXt-Tiny, EfficientNet-B3, DINOv2, and DINOv3 use up to 400 epochs with patience 50; the minimum improvement is 10^{-4} . Depths 0, 1, and 2 share one partition per backbone, and the preferred depth and checkpoint are chosen by validation accuracy only; test results never drive selection. The setting is *transductive*: validation and test leaf nodes are in the graph and take part in message passing, but their labels are excluded from the loss and they have no has_disease edges. Results therefore measure classification of unlabeled nodes already in the graph, not fully inductive prediction for images added after training.

I. Reference Models and Ablation Design

Logistic-regression reference. A multinomial logistic regression (up to 2,000 iterations) is trained per backbone on the same standardized features without any graph, crop, disease, or category nodes, HGT, fusion, or decoder, reflecting conventional feature-only recognition [1], [2]. It uses all 25,122

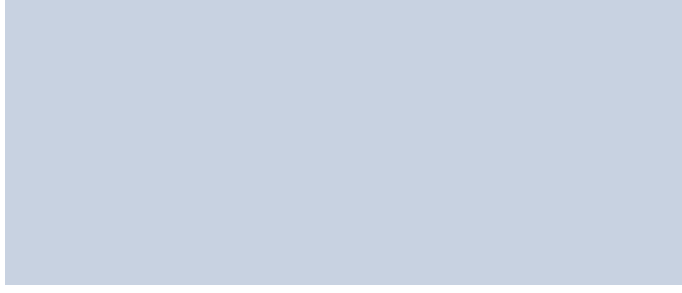


Fig. 4. AgriHGT architecture: visual feature extraction, heterogeneous graph construction, relation-aware HGT message passing, residual visual fusion, and disease prediction.

training labels, since no label-bearing edges require a message/supervision split, so it is a visual-feature reference rather than a matched supervision ablation. **Zero-layer reference.** HGT-0 keeps the rest of AgriHGT, so comparing it with HGT-1 and HGT-2 isolates message propagation.

Component ablation. With DINOv3 features, two HGT layers (best DINOv3 validation depth) form the full reference A0, and one component is changed at a time: A1 zero HGT layers; A2 no `similar_to`; A3 no crop nodes, `grown_on`, or `affects`; A4 no category nodes, `is_type_of`, or `co_category`; A5 no `has_disease`; A6 zero meta-node embeddings; A7 no residual visual bypass; and A8 a fixed equal-weight average instead of the gate (all with reverses). We keep hidden size 256, 4 heads, dropout 0.2, the MLP decoder, the weighted loss, the split, the feature bank, the `train_msg/train_sup` partition, and the validation and test sets. For A2–A5 the graph is rebuilt without only the named nodes or relations (the within-crop neighbor graph is kept for A3–A5 and removed only in A2). A6 keeps the structure but zeroes the meta-node embeddings, A7 feeds $\mathbf{h}_i$ directly to the decoder, and A8 replaces Eq. (2) with $\mathbf{z}_i = 0.5\mathbf{r}_i + 0.5\mathbf{h}_i$. A0 and A2–A8 use up to 400 epochs, AdamW (10^{-3} , 10^{-4}), cosine decay to 10^{-6} , patience 50, minimum improvement 10^{-4} , and seed 42; A1 reuses the zero-layer DINOv3 run under the same settings. With a single seed, differences are component-level observations, which the paired McNemar tests below qualify.

J. Evaluation Metrics

For N test images with per-class counts TP_c, FP_c, FN_c, TN_c , we report accuracy, precision $P_c = TP_c / (TP_c + FP_c)$, recall $R_c = TP_c / (TP_c + FN_c)$, specificity $Sp_c = TN_c / (TN_c + FP_c)$, $F1_c = 2P_cR_c / (P_c + R_c)$, and Jaccard $J_c = TP_c / (TP_c + FP_c + FN_c)$, with macro averages $\frac{1}{28} \sum_c (\cdot)$ that weight every class equally; balanced accuracy is the macro recall. Chance-corrected agreement uses Cohen’s $\kappa = (p_o - p_e) / (1 - p_e)$, $p_e = \sum_c n_c \hat{n}_c / N^2$, and multiclass $MCC = (CN - \sum_c \hat{n}_c n_c) / \sqrt{(N^2 - \sum_c \hat{n}_c^2)(N^2 - \sum_c n_c^2)}$, where C is the number correct and $n_c, \hat{n}_c$ are true and predicted counts. From predicted probabilities we report top-3 accuracy, macro one-vs-rest ROC-AUC, negative log-likelihood (NLL), and expected calibration error $ECE = \sum_b \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)|$ over 15 confidence bins. Accuracy has a 95% Wilson interval

$$\left(\hat{p} + \frac{z^2}{2N} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}} \right) / \left(1 + \frac{z^2}{N} \right), \quad z = 1.96, \quad (5)$$

macro-F1 a 95% bootstrap interval (2,000 resamples of the test set), and each ablation is compared with A0 on the same 848 images by an exact McNemar test on the discordant pairs (n_{01}, n_{10}) , $p = 2 \sum_{k \leq \min(n_{01}, n_{10})} \binom{n_{01} + n_{10}}{k} 2^{-(n_{01} + n_{10})}$. All values are computed by script directly from saved predictions.

K. Experimental Setup

Experiments ran on Kaggle with one NVIDIA Tesla T4 GPU. We used PyTorch and Torchvision (backbones), the official DINOv2 code and `timm` DINOv3, `pyHGT` (with the zero-layer change) and PyTorch Geometric (graphs), `scikit-learn` (neighbors, logistic regression, evaluation), `Pandas` and `NumPy`, and `Pillow` and `ImageHash` (perceptual hashing). The split was fixed across backbones, seed 42 was used for the main and ablation experiments, and a fixed hold-out validation set (not k -fold cross-validation) selected checkpoints and depths. All experiments are single-seed; multi-seed variability was not estimated and is a limitation.

IV. RESULTS

Backbones and depth. Table III reports every backbone with the feature-only reference, HGT depths 0–2, and the validation-selected depth. The strongest configuration is [backbone, depth], with [xx.xx]% test accuracy (95% CI [[xx.xx, xx.xx]]), [0.xxxx] macro-F1, and $\kappa = [0.xxxx]$. Relative to its logistic-regression reference, message passing changes accuracy by [$\pm$ x.xx] points and macro-F1 by [$\pm$ 0.xxxx]; across backbones, HGT-1/2 improves on HGT-0 for [k] of 7 backbones. Self-supervised DINO features [do / do not] outperform the supervised CNN and Swin features.

Component ablation. Table IV shows that removing [component] causes the largest macro-F1 drop ([−0.xxxx], McNemar $p = [0.xxx]$), while [components] change results by less than [x] points and are not significant at $p < 0.05$.

Per-class behavior. Per-class precision, recall, specificity, F1, Jaccard, and MCC (supplementary table, generated by the same script) show the lowest F1 for [classes], mostly minority classes, whereas [classes] are recognized almost perfectly.

V. DISCUSSION AND LIMITATIONS

HGT-0 differs from HGT-1/2 only in propagation, so it is the direct test of message passing; logistic regression shows what the adapted features already hold but uses all 25,122 training labels against 6,241 `train_sup` nodes, so it is not a matched

TABLE III
TEST RESULTS PER BACKBONE ($N = 848$): ACCURACY (%) AND MACRO-F1 OF THE FEATURE-ONLY LOGISTIC REGRESSION (LR) AND AGRIHGT WITH 0–2 HGT LAYERS, AND FULL METRICS FOR THE DEPTH L^* SELECTED BY VALIDATION ACCURACY

Backbone	LR		HGT-0		HGT-1		HGT-2		L^*	Selected depth L^*					
	Acc	F1	Acc	F1	Acc	F1	Acc	F1		95% CI	Bal. Acc	κ	MCC	AUC	
ResNet50	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
DenseNet121	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
EfficientNet-B3	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
ConvNeXt-Tiny	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Swin-Tiny	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
DINOv2 ViT-B/14	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
DINOv3 ViT-B/16	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–

TABLE IV
COMPONENT ABLATION WITH DINOv3 FEATURES ($N = 848$). Δ RELATIVE TO A0; MACRO-F1 CI BY BOOTSTRAP; p FROM THE EXACT MCNEMAR TEST AGAINST A0

ID	Configuration	Val Acc	Test Acc	Δ Acc	Macro-F1	95% CI	Δ F1	Bal. Acc	κ	MCC	NLL	ECE	McNemar p
A0	Full AgriHGT	–	–	–	–	–	–	–	–	–	–	–	–
A1	No HGT message passing	–	–	–	–	–	–	–	–	–	–	–	–
A2	No leaf similarity	–	–	–	–	–	–	–	–	–	–	–	–
A3	No crop knowledge	–	–	–	–	–	–	–	–	–	–	–	–
A4	No disease taxonomy	–	–	–	–	–	–	–	–	–	–	–	–
A5	No visual grounding	–	–	–	–	–	–	–	–	–	–	–	–
A6	No meta-node identity	–	–	–	–	–	–	–	–	–	–	–	–
A7	No residual bypass	–	–	–	–	–	–	–	–	–	–	–	–
A8	No fusion gate	–	–	–	–	–	–	–	–	–	–	–	–

ablation. The gate initialization ($\sigma(2) \approx 0.88$) lets the model fall back on visual evidence, which makes A7 informative about what the graph alone carries. Limitations: evaluation is transductive, so images added after training are untested; leakage control covers same-class pairs within Hamming distance six only; one seed and split mean differences inside the Wilson and bootstrap intervals should not be over-read; and the crops come from one region.

VI. CONCLUSION

AgriHGT combines backbone features of leaf images with crop, disease, and disease-category knowledge in a heterogeneous graph, processed by relation-aware HGT message passing, gated residual fusion, and a compatibility decoder. On a leakage-controlled split of Agri-Vision4 with a label-shortcut-free supervision scheme, it was evaluated across seven backbones, three depths, and nine ablations, reaching [xx.xx]% accuracy and [0.xxxx] macro-F1 with [backbone]. Future work will evaluate inductive inference, multiple seeds and splits, cross-class near-duplicate auditing, and additional crops and regions.

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